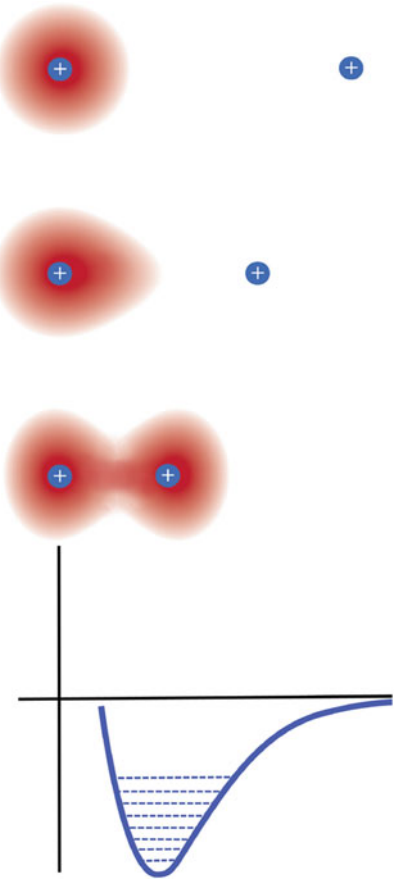


Fig. 1.4 The H_2^+ interaction potential (charge cloud shown for illustration purposes only)

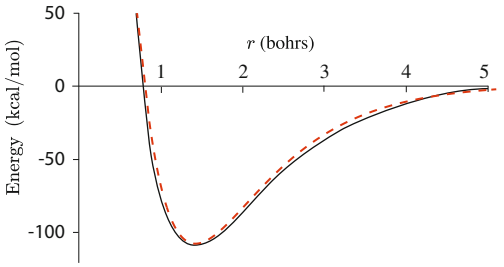


Fig. 1.5 The H_2 bond potential (*solid*) with a Morse potential fitted to it (*dashed curve*). Reference H_2 potential curve from [351]

Because they often do not need to be allowed to break during simulation and are very strong compared to other potential terms, *chemical bonds* such as the covalent H_2^+ bond described above are sometimes treated as springs with given rest-length:

$$\varphi_{ij}^{\text{len}}(r_{ij}) = \frac{k_{ij}^{\text{len}}}{2}(r_{ij} - r_{ij}^0)^2, \qquad r_{ij} = \|\boldsymbol{q}_i - \boldsymbol{q}_j\|.$$